

WATERTOWN, NY, USA

WMO: 726227

Lat: 43.989N

Lon: 76.026W

Elev: 318

StdP: 14.53

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94790

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-5.4	-19.9	1.9	-12.0	-12.8	2.8	-4.4	28.3	28.0	26.1	29.1	4.3	80	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	85.7	71.8	83.3	70.7	81.1	69.2	74.6	82.3	73.0	80.2	71.4	78.2	10.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.1	120.1	79.1	70.4	113.1	77.6	68.8	107.2	76.4	38.2	82.8	36.7	80.1	35.4	78.2	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.2	19.6	17.7	DB	-27.4	90.1	6.6	2.6	-32.2	91.9	-36.0	93.5	-39.7	94.9	-44.5	96.8	(4)
(5)			WB	-27.3	77.2	6.2	1.7	-31.8	78.4	-35.4	79.5	-38.9	80.4	-43.3	81.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.0	20.2	21.8	30.9	43.5	55.5	64.4	69.4	67.9	60.7	49.6	38.4	27.8
	DBStd	19.73	14.88	12.66	11.38	8.84	7.86	6.70	5.37	5.70	7.68	8.67	9.66	11.57
	HDD50	3761	927	791	599	234	33	0	0	0	7	116	365	689
	HDD65	7366	1390	1210	1058	646	312	92	19	36	172	482	797	1152
	CDD50	2292	2	1	6	40	202	431	602	556	329	104	17	2
	CDD65	422	0	0	0	2	16	73	156	127	44	4	0	0
	CDH74	3406	0	0	3	30	176	600	1266	968	342	21	0	0
	CDH80	695	0	0	0	5	25	129	272	197	66	1	0	0
Wind	WSAvg	7.9	9.2	8.9	8.9	9.0	7.6	6.9	6.8	6.3	6.6	8.0	8.5	8.7
Precipitation	PrecAvg	32.1	2.4	1.9	2.4	2.4	2.8	2.6	2.1	3.1	3.4	2.8	3.5	2.8
	PrecMax	42.4	7.4	5.5	5.7	5.3	5.6	6.2	4.7	7.7	7.4	6.1	7.2	6.7
	PrecMin	20.8	0.5	0.2	0.2	0.6	0.7	0.3	0.3	0.9	0.9	0.2	1.1	0.3
	PrecStd	5.5	1.7	1.1	1.2	1.3	1.3	1.4	1.2	1.4	1.5	1.3	1.6	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.0	56.3	68.3	78.3	83.3	87.7	88.7	88.3	86.2	76.9	67.6	60.5
		MCWB	55.1	48.0	57.6	61.2	68.3	72.7	74.5	72.6	71.9	64.4	57.5	55.1
	2%	DB	51.1	49.0	58.6	70.7	79.0	83.8	85.4	84.6	81.6	72.1	62.9	52.0
		MCWB	47.6	43.2	49.4	56.9	65.8	70.3	72.2	71.2	70.1	62.8	54.0	47.5
	5%	DB	44.4	43.9	53.3	64.5	74.9	80.5	83.1	82.0	77.8	67.9	58.3	47.8
		MCWB	40.7	39.1	46.3	52.4	63.1	68.3	71.2	69.8	67.9	60.0	52.1	44.0
	10%	DB	38.9	38.9	47.5	59.4	70.9	77.2	80.8	79.4	74.6	63.9	53.7	43.1
		MCWB	35.7	35.2	41.5	49.9	59.9	66.7	69.7	68.8	65.8	56.8	47.9	39.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.8	49.9	58.1	63.7	71.1	75.0	77.3	76.3	74.4	68.1	59.5	54.7
		MCDB	59.7	55.2	65.8	73.7	80.6	84.1	85.5	84.2	82.2	73.0	64.9	59.6
	2%	WB	47.5	44.0	51.3	58.8	67.9	72.5	75.0	73.8	71.6	64.2	56.0	48.3
		MCDB	50.7	48.3	56.8	67.8	76.4	80.8	82.6	81.0	79.0	70.4	61.1	50.8
	5%	WB	41.0	39.3	46.4	54.8	65.0	70.3	73.3	72.2	69.3	60.6	52.4	44.1
		MCDB	43.8	43.5	52.5	62.2	72.5	77.6	80.3	78.8	75.9	66.6	57.5	47.5
	10%	WB	36.0	35.6	41.8	50.7	62.0	68.4	71.7	70.5	67.2	57.9	48.4	39.5
		MCDB	38.7	38.5	46.9	58.3	68.9	75.0	78.3	76.8	73.2	63.5	53.1	42.9
Mean Daily Temperature Range	5% DB	MDBR	18.6	19.2	18.2	20.1	21.0	19.8	19.1	20.0	20.9	17.8	16.0	15.2
		MCDBR	18.7	20.7	25.1	29.5	26.1	23.2	21.2	22.2	22.7	21.8	22.0	18.3
		MCWBR	17.0	16.8	17.6	18.1	15.0	12.9	11.6	11.8	13.1	14.3	16.1	15.9
	5% WB	MCDBR	18.0	20.1	22.8	25.6	22.4	19.2	17.2	17.9	19.7	19.6	19.9	17.5
		MCWBR	16.9	16.8	17.0	17.4	14.2	11.6	10.7	10.9	12.2	13.8	15.9	15.9
Clear-Sky Solar Irradiance	taub	0.295	0.299	0.328	0.359	0.387	0.421	0.430	0.420	0.373	0.353	0.312	0.287	
	taud	2.388	2.375	2.404	2.385	2.358	2.288	2.265	2.288	2.406	2.451	2.510	2.475	
	Ebn at Noon	264	284	289	287	280	270	266	265	270	261	257	257	
	Edh at Noon	24	29	33	36	38	41	42	39	32	27	22	20	
All-Sky Solar Radiation	RadAvg	412	682	1094	1401	1696	1807	1846	1633	1299	770	480	318	
	RadStd	45	70	92	92	165	171	147	159	81	112	74	57	42

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (30 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page